KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

NAME:

TEACHER:

MATHEMATICS

Year 8 – Practice Examination 3

Equations, Investigating Data and Mixed Topics

Practice paper for the Year 8 Yearly Examination

Instructions		
1. Attempt all questions 2. Show working where needed 3. Time Allowed 60 minutes 4. Calculators are permitted 5. Marks are shown beside each question	Equations (Section A):	/18
	Investigating Data (Section B):	/21
	Mixed Topics (Section C):	/21
	Total Marks:	/60

Section A – Equations (18 marks)

Suggested time: 20 minutes. Show all working.

Question 1 – One-step equations **(4 marks)**

Solve each equation.

a) $x + 19 = 33$ x = _____ **(1 mark)**

Working: _____

b) $m - 11 = 23$ m = _____ **(1 mark)**

Working: _____

c) $8p = 72$ p = _____ **(1 mark)**

Working: _____

d) $y \div 3 = 11$ y = _____ **(1 mark)**

Working: _____

Question 2 – Two-step equations **(4 marks)**

Solve each equation, showing your working.

a) $7x + 9 = 51$ **(2 marks)**

Working: _____

_____

_____

x = _____

b) $x \div 4 - 3 = 5$ **(2 marks)**

Working: _____

_____

_____

x = _____

Question 3 – Equations with a negative solution **(2 marks)**

Solve $5m + 32 = 2$.

Working: _____

_____

_____

m = _____

Question 4 – Checking solutions **(2 marks)**

Use substitution to show whether $x = 7$ is the solution to $5x - 8 = 26$.

Working: _____

_____

_____

_____

Is x = 7 the solution? (circle one) YES / NO

Question 5 – Worded problem **(3 marks)**

I think of a number, multiply it by 6, and then add 11. The answer is 59.

a) Write an equation to represent this problem. Use *n* for the number. **(1 mark)**

Equation: _____

b) Solve your equation to find the number. **(2 marks)**

Working: _____

_____

_____

n = _____

Question 6 – Worded problem **(3 marks)**

A school excursion costs $65 for the bus plus $9 for each student who goes. The total cost is $290. Let *s* be the number of students who go on the excursion.

a) Write an equation to represent this situation. **(1 mark)**

Equation: _____

b) Solve your equation to find how many students went on the excursion. **(2 marks)**

Working: _____

_____

_____

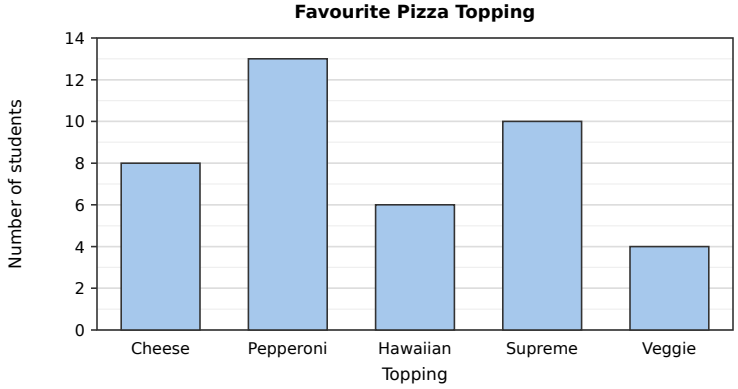
s = _____

Section B – Investigating Data (21 marks)

Suggested time: 20 minutes. Show all working.

Question 7 – Column graph **(2 marks)**

Year 8 students voted for their favourite pizza topping. The results are shown below.



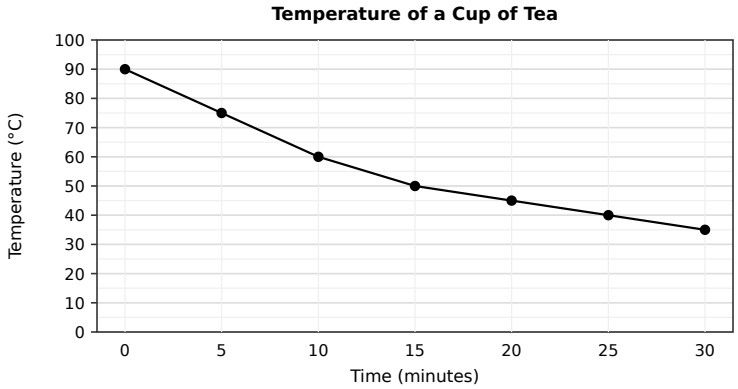
a) Which topping was the most popular? _____ **(1 mark)**

b) How many students voted altogether? _____ **(1 mark)**

Working: _____

Question 8 – Line graph **(2 marks)**

The line graph shows the temperature of a cup of tea as it cools down.



a) After how many minutes was the temperature 50°C? _____ **(1 mark)**

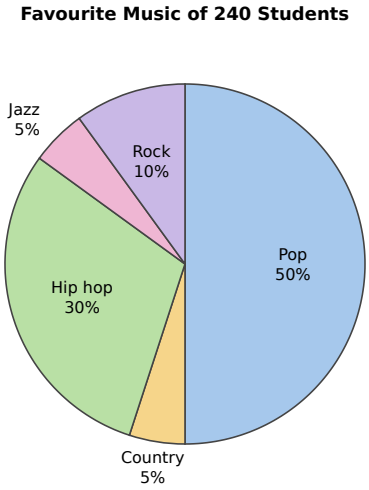
b) By how many degrees did the temperature fall between 0 and 20 minutes? **(1 mark)**

Working: _____

Answer: _____

Question 9 – Sector (pie) graph **(3 marks)**

240 students were asked what type of music they like best. The results are shown in the sector graph.



a) How many students chose hip hop? **(1 mark)**

Working: _____

Answer: _____

b) How many more students chose pop than rock? **(2 marks)**

Working: _____

_____

_____

Answer: _____

Question 10 – Stem-and-leaf plot **(2 marks)**

The number of push-ups that 12 students did in one minute is shown in the stem-and-leaf plot.

Stem	Leaf
1	6 9
2	1 3 5 5 8
3	0 2 4 7
4	3

Key: 2 | 1 means 21

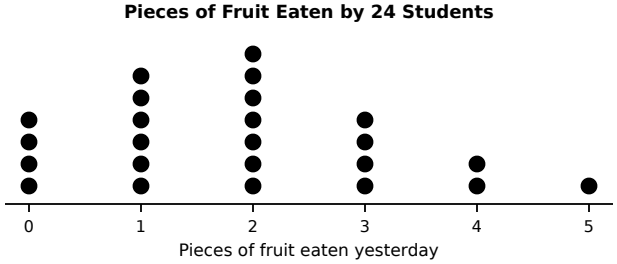
a) Find the median. _____ **(1 mark)**

Working: _____

b) How many students did more than 25 push-ups? _____ **(1 mark)**

Question 11 – Dot plot **(2 marks)**

Twenty-four students were asked how many pieces of fruit they ate yesterday. The results are shown in the dot plot.



a) What is the mode? _____ **(1 mark)**

b) How many students ate at least 3 pieces of fruit? _____ **(1 mark)**

Question 12 – Frequency table **(2 marks)**

Students in Year 8 were asked how many brothers and sisters they have. The results are shown in the frequency table.

Brothers and sisters	0	1	2	3	4
Frequency	7	13	8	3	1

a) How many students were surveyed altogether? _____ **(1 mark)**

Working: _____

b) What is the mode? _____ **(1 mark)**

Question 13 – Mean, median, mode and range **(5 marks)**

The number of hours that ten students spent on a screen yesterday is shown below.

5, 8, 3, 7, 8, 10, 6, 8, 4, 11

a) Calculate the mean. **(2 marks)**

Working: _____

_____

_____

Mean = _____

b) Find the median. _____ **(1 mark)**

Working: _____

c) Find the mode. _____ **(1 mark)**

d) Find the range. _____ **(1 mark)**

Working: _____

Question 14 – Outliers **(3 marks)**

The prices, in dollars, of six second-hand phones are shown below.

120, 135, 110, 460, 125, 130

a) Identify the outlier in this data set. _____ **(1 mark)**

b) Calculate the mean of the data, including the outlier. **(1 mark)**

Working: _____

_____

Mean = _____

c) Does the outlier make the mean higher or lower than most of the data values? Explain your answer. **(1 mark)**

Working: _____

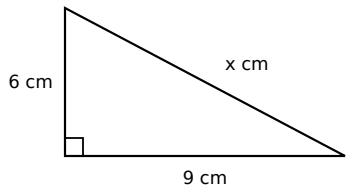
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Section C – Mixed Topics (21 marks)

Suggested time: 20 minutes. Show all working.

Question 15 – Pythagoras' Theorem **(4 marks)**

a) Find the length of the hypotenuse, *x*, in the triangle below. Give your answer correct to one decimal place. **(2 marks)**



Working: _____

_____

_____

x = _____

b) A triangle has side lengths 9 cm, 40 cm and 41 cm. Is it a right-angled triangle? Show working. **(2 marks)**

Working: _____

_____

_____

_____

Question 16 – Decimals **(2 marks)**

a) Calculate 1.25 × 0.4 _____ **(1 mark)**

Working: _____

_____

b) Calculate 4.2 ÷ 0.7 _____ **(1 mark)**

Working: _____

_____

Question 17 – Powers and roots **(2 marks)**

a) Evaluate 10^3 _____ **(1 mark)**

Working: _____

b) Evaluate √225 _____ **(1 mark)**

Working: _____

Question 18 – Surds **(3 marks)**

a) Circle the surds: √27 √81 √14 √64 **(1 mark)**

b) Simplify √75 **(2 marks)**

Working: _____

_____

_____

Answer: _____

Question 19 – Algebra **(3 marks)**

a) Simplify 9k + 5 − 4k + 3 _____ **(1 mark)**

Working: _____

b) Expand 6(2x + 1) _____ **(1 mark)**

Working: _____

c) Evaluate $5t^2 - 2u$ when t = 2 and u = 7. _____ **(1 mark)**

Working: _____

_____

Question 20 – Area and volume **(3 marks)**

a) Find the area of a trapezium with parallel sides 11 cm and 7 cm and height 4 cm. **(2 marks)**

Formula: Area = ½ × (a + b) × h, where a and b are the parallel sides and h is the perpendicular height.

Working: _____

_____

_____

Answer: _____

b) Find the volume of a rectangular prism that is 6 cm long, 6 cm wide and 10 cm high. **(1 mark)**

Working: _____

_____

Answer: _____

Question 21 – Linear relationships **(2 marks)**

Complete the table of values for the rule y = 5x + 2.

x	−2	−1	0	1
y				

Question 22 – Rates and ratios **(2 marks)**

a) Write the ratio 28 : 42 in simplest form. _____ **(1 mark)**

Working: _____

_____

b) A runner runs 30 km in 2.5 hours. Find the runner's average speed in km/h. **(1 mark)**

Working: _____

_____

Answer: _____

END OF EXAMINATION PAPER